

A Scalable Cloud-Based Framework for Adaptive Traffic Signal Control Utilizing Deep Reinforcement Learning in Urban Environments

Zainab Saadoon Naser^{1,2}, Hend Marouane², Ahmed Fakhfakh^{1,3}, Zaid Abdulsalam Almatwari⁴

¹National School of Electronics and Telecommunications (ENET'COM)

University of Sfax, Sfax 3021, Tunisia

²National School of Electronics and Telecommunications (ENET'COM), NTS'COM Laboratory

University of Sfax, Sfax 3029, Tunisia

³Laboratory of Signals, Systems, Artificial Intelligence and Networks (SM@RTS)

Digital Research Center of Sfax (CRNS), Sfax 3021, Tunisia

⁴Department of Information and Communications Engineering

Al Farqadein University, Basra, Iraq

Abstract—Efficient traffic signal control is a cornerstone of modern urban mobility, playing a critical role in reducing congestion, travel delays, and emissions. Traditional fixed-time or rule-based traffic signal systems often lack the flexibility to adapt to dynamic and non-uniform traffic patterns, particularly in densely populated cities. To address these limitations, this paper proposes a scalable, cloud-based framework for adaptive traffic signal optimization using multi-agent deep reinforcement learning (DRL). In the proposed system, each intersection is governed by an independent agent trained with the Proximal Policy Optimization (PPO) algorithm. Agents learn traffic signal policies from localized, real-time observations, including vehicle counts, waiting times, and signal states, enabling decentralized, context-aware control. The framework is designed for cloud deployment, leveraging Google Colab and Google Cloud Storage for scalable training, consistent metric logging, and reproducible post-simulation analysis. Traffic behavior is modeled using the Simulation of Urban Mobility (SUMO) platform, ensuring a realistic and flexible environment for evaluation. Extensive experiments demonstrate that the DRL-based system significantly outperforms traditional fixed-time control across key metrics such as average waiting time, vehicle queue length, throughput, and congestion levels. Performance improvements are especially evident under high-density and variable traffic conditions, where adaptability is most critical. In addition, the system exhibits robust convergence across episodes, while the cloud-integrated design ensures scalability, reproducibility, and practical applicability for smart city deployment.

Index Terms—Adaptive traffic signal control, cloud computing, deep reinforcement learning, multi-agent systems, real-time traffic optimization, scalable architecture

I. INTRODUCTION

As urban populations continue to expand, cities face mounting challenges in mitigating traffic congestion, reducing travel delays, and minimizing emissions. Conventional traffic signal control strategies, such as fixed-time and actuated signal plans, are increasingly inadequate for adapting to the highly dynamic and non-uniform nature of modern traffic conditions, often leading to inefficiencies and commuter dissatisfaction [1], [2].

In response, the transportation research community has increasingly turned toward intelligent traffic control solutions powered by machine learning, with Deep Reinforcement Learning (DRL) showing particular promise due to its ability

to learn adaptive control policies directly from interactions with complex, dynamic environments [3]. While DRL-based methods can optimize key metrics such as average waiting time, queue length, and throughput, extending these methods to multi-intersection networks requires decentralized coordination, which has been explored through Multi-Agent Reinforcement Learning (MARL) [2], [4].

At the same time, cloud computing has emerged as a critical enabler of scalable and reproducible intelligent transportation systems by providing the computational resources required to train DRL agents, manage high-fidelity simulations, and support centralized monitoring and logging [5], [6]. Building on these developments, this paper introduces a cloud-based multi-agent DRL framework for adaptive traffic signal control that integrates SUMO simulation with agents trained via the Proximal Policy Optimization (PPO) algorithm, deployed on Google Colab and Google Cloud Storage (GCS). Each intersection is modeled as an independent agent that observes localized traffic states and makes real-time signal control decisions, while the cloud infrastructure ensures scalability, reproducibility, and ease of deployment in smart city contexts. Experimental results demonstrate that the proposed system significantly outperforms traditional fixed-time baselines across multiple performance indicators, including average waiting time, vehicle queue length, traffic flow, and congestion level, particularly under variable and high-density traffic conditions, thereby underscoring its potential as a scalable and practical solution for future urban mobility systems.

II. BACKGROUND

Traffic signal control plays a pivotal role in urban transportation systems, serving as a mechanism to regulate the flow of vehicles and pedestrians at intersections. Traditionally, traffic signals have operated using fixed-time or actuated control strategies, which are configured based on historical traffic patterns or sensor-triggered inputs. While effective in static or predictable environments, these approaches often lack the flexibility to adapt to real-time variations in traffic flow, leading to inefficiencies and congestion in dynamic urban settings [7].

To overcome these limitations, the field has increasingly adopted data-driven and adaptive control techniques. Among these, Deep Reinforcement Learning (DRL) has emerged as a

powerful framework for modeling complex decision-making environments. In DRL, agents learn policies through interaction with the environment, receiving rewards based on the effectiveness of their actions. Applied to traffic control, DRL allows agents to determine optimal signal phases by continuously adjusting to real-time traffic conditions, thereby improving flow and reducing delay [8].

The use of Multi-Agent Reinforcement Learning (MARL) extends these benefits to networks of intersections. Rather than relying on a centralized controller, MARL enables each intersection to act as an autonomous agent, learning and adapting to its local environment while optionally sharing information with neighboring agents [4], [9]. This decentralized architecture enhances system scalability, resilience, and coordination in large-scale traffic networks.

In parallel with algorithmic advancements, cloud computing technologies have transformed how intelligent transportation systems are designed and deployed. Cloud platforms provide the necessary computational resources to train large-scale DRL models, manage traffic simulations, and store high-volume traffic data. Real-time data ingestion and monitoring are also facilitated through cloud services, enabling adaptive signal control systems to respond dynamically to changing urban conditions [10].

Additionally, simulation environments such as the Simulation of Urban MObility (SUMO) have become indispensable tools in the development and evaluation of intelligent traffic control systems. SUMO offers a flexible, open-source platform for modeling road networks, traffic flow, and signal behavior, making it suitable for training and testing DRL-based agents in controlled yet realistic settings [11].

III. RELATED WORK

Recent years have seen significant progress at the intersection of intelligent transportation systems, machine learning, and cloud computing. Existing research can broadly be grouped into four areas: traditional and machine learning-based traffic signal control, reinforcement learning for traffic management, multi-agent reinforcement learning for urban networks, and the integration of cloud-based infrastructures into intelligent traffic systems.

Traditional traffic signal optimization has relied on heuristics, mathematical programming, and rule-based adaptive strategies. While these methods provide stability in predictable traffic environments, their effectiveness declines under stochastic or rapidly changing conditions [1], [12]. To overcome these limitations, machine learning models, including supervised and unsupervised techniques, have been applied to tasks such as traffic flow prediction and congestion detection [13], [14]. Although effective for forecasting, these approaches are not inherently designed for sequential decision-making in control settings.

Reinforcement learning (RL) has emerged as a promising alternative, allowing agents to optimize traffic signals through interaction with simulated environments. Deep Reinforcement Learning (DRL) approaches, in particular, have demonstrated substantial improvements in reducing vehicle waiting times and improving throughput at intersections [15]. Various architectures, such as Deep Q-Networks (DQNs), Actor-Critic models, and Proximal Policy System Architecture Optimization (PPO), have been

successfully applied to single and multi-phase traffic signal control problems.

Building on these successes, Multi-Agent Reinforcement Learning (MARL) extends RL to networks of intersections, where each signal controller is modeled as an autonomous agent. MARL systems have been shown to improve scalability and resilience compared to centralized approaches, while supporting coordination through inter-agent communication or shared global objectives. Recent works have explored cooperative and competitive MARL frameworks [16], as well as graph neural network (GNN)-based methods for capturing spatial relationships among intersections [17]. However, many of these frameworks remain limited in scalability or lack deployment strategies suitable for real-world, cloud-enabled infrastructures.

In parallel, researchers have explored the integration of cloud computing into traffic management systems. Cloud platforms provide scalable training resources, centralized logging, and real-time data ingestion, enabling DRL-based traffic control to operate at urban scales [18]. Combined with simulation tools such as the Simulation of Urban MObility (SUMO), cloud architectures allow for reproducible and high-fidelity experimentation [19].

Despite these advances, few studies fully integrate MARL-based traffic control with cloud-native training pipelines and large-scale SUMO simulations. Existing works often either focus on algorithmic contributions without deployment considerations, or on cloud-based traffic prediction without decentralized RL optimization. This paper addresses this gap by presenting a cloud-deployable MARL framework that unifies scalable training, SUMO-based simulation, and real-time performance monitoring, thereby advancing both the methodological and infrastructural aspects of adaptive traffic signal control.

As illustrated in Table I, previous studies have addressed individual aspects of traffic optimization, such as simulation-based control, adaptive learning, or cloud architecture, but few integrate all five pillars into a unified, deployable framework. Our work uniquely combines multi-agent DRL with SUMO-based simulation and real-time cloud-compatible performance tracking, offering a more complete and scalable solution for adaptive urban traffic signal control.

IV. SYSTEM ARCHITECTURE

The proposed architecture enables scalable and adaptive traffic signal control through a cloud-integrated, multi-agent deep reinforcement learning (DRL) framework. It consists of four key modules: (i) a high-fidelity simulation environment, (ii) a decentralized DRL control module, (iii) a cloud-based training and logging infrastructure, and (iv) a real-time evaluation engine. The following subsections describe each component and the flow of information between them.

The traffic network is modeled using the Simulation of Urban Mobility (SUMO), a microscopic simulator capable of representing vehicle interactions, lane changes, and signal phases. SUMO provides the environment state space for the learning agents, including lane-specific queue lengths, average waiting times, and traffic densities across intersections. These features form the basis of the agents

TABLE I
COMPARISON OF OUR WORK WITH EXISTING TRAFFIC SIGNAL CONTROL APPROACHES

Study	Multi-Agent DRL	Cloud Integration	SUMO Simulation	Real-Time Metrics	Adaptive Control
Ahmadi et al. [2]	✓	✗	✓	✗	✓
Michailidis et al. [7]	✗	✗	✓	✓	✓
Li et al. [4]	✓	✗	✓	✓	✓
Haiyan et al. [20]	✓	✗	✓	✗	✓
Wan et al. [21]	✗	✗	✗	✗	✓
Moumen et al. [11]	✗	✗	✓	✓	✗
Amit et al. [22]	✗	✗	✓	✓	✓
Sharma et al. [17]	✓	✗	✗	✓	✓
Bellavista et al. [5]	✗	✓	✗	✗	✗
Abbasi et al. [23]	✗	✓	✗	✗	✗
Our Work	✓	✓	✓	✓	✓

observation vectors, enabling data-driven decision-making under realistic traffic dynamics. Each intersection is managed by an autonomous agent trained with the Proximal Policy Optimization (PPO) algorithm. The control objective is to maximize the expected cumulative discounted reward:

$$J(\theta) = E_{\pi_{\theta}} \left[\sum_{t=0}^T \gamma^t r_t \right] \quad (1)$$

where π_{θ} denotes the policy parameterized by θ , r_t is the reward at time t , and $\gamma \in (0, 1]$ is the discount factor.

To ensure stable updates, PPO employs a clipped surrogate objective:

$$L_{\text{CLIP}}(\theta) = E_t \left[\min \left(\frac{r_t(\theta) \widehat{A}_t}{\text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \widehat{A}_t} \right) \right] \quad (2)$$

where $r_t(\theta)$ is the probability ratio between the new and old localized rewards r_t from SUMO. Once an episode concludes, policies, and $\widehat{A}_t$ represents the estimated advantage function.

Training and evaluation are executed in a cloud environment using Google Colab for compute resources and Google Cloud Storage (GCS) for centralized logging. Cloud integration enables scalable training cycles, consistent experiment tracking, and remote reproducibility. At the end of each simulation episode, traffic performance metrics are computed as follows:

$$\text{Average Waiting Time} = \frac{1}{N} \sum_{i=1}^N T_i \quad (3)$$

$$\text{Traffic Flow} = \frac{\text{Total Vehicles}}{\text{Time Interval}} \quad (4)$$

$$\text{Queue Length} = \frac{\text{Vehicle Lengths}}{\text{Number of Vehicles}} \quad (5)$$

$$\text{Congestion Level} = \frac{\text{Actual Flow}}{\text{Ideal Flow}} \quad (6)$$

Here, N denotes the number of vehicles, and T_i is the waiting time of vehicle i . These metrics are aggregated across intersections and simulation runs to ensure consistency in evaluation.

The control strategy builds on a decentralized multi-agent variant of PPO. Each intersection is treated as an independent agent A_i , parameterized by a local policy π_{θ_i} and value function V_{ϕ_i} . At each timestep, agents observe their local

states s_i^t , select actions a_i^t (signal phase changes), and receive localized rewards r_i^t from SUMO. Once an episode concludes, agents compute advantage estimates $\hat{A}_i^t$ using Generalized Advantage Estimation (GAE), update their policy parameters by optimizing L^{CLIP} , and refine their value function by minimizing the temporal-difference loss L^{VF} . The complete training process is summarized in Algorithm 1. The roles of the main components are summarized in Table II.

Algorithm 1 Multi Agent PPO for Traffic Signal Control

```

1: for each agent  $A_i$  do
2:   Initialize policy  $\pi_{\theta_i}$  and value function  $V_{\phi_i}$ 
3: end for
4: for each training episode do
5:   for each timestep  $t$  do
6:     for each agent  $A_i$  do
7:       Observe  $s_i^t$ , choose  $a_i^t \sim \pi_{\theta_i}(\cdot | s_i^t)$ 
8:       Apply  $a_i^t$  in SUMO, receive  $r_i^t, s_i^{t+1}$ 
9:     end for
10:   end for
11:   for each agent  $A_i$  do
12:     Estimate  $\hat{A}_i^t$  and update  $\theta_i$  with  $L^{\text{CLIP}}$ 
13:     Update  $\phi_i$  with  $L^{VF}$ 
14:   end for
15: end for

```

TABLE II
SYSTEM MODULES AND RESPONSIBILITIES

Component	Functionality
Traffic Simulation (SUMO)	Generates intersection-level states including vehicle queues, waiting times, and densities.
Multi-Agent PPO Agents	Learn decentralized signal control policies based on local traffic observations.
Cloud Infrastructure (Colab)	Provides scalable compute for DRL training and inter-agent co-ordination.
Metric Logger (GCS)	Stores performance metrics and simulation logs for reproducibility.
Evaluation Engine	Aggregates results across episodes and computes key performance indicators.

V. METHODOLOGY

This section describes the methodology used to design, implement, and evaluate the proposed cloud-based multi-agent deep reinforcement learning (DRL) framework for adaptive traffic signal control. The workflow consists of four main stages: traffic environment modeling, agent design, training protocol, and performance evaluation.

The traffic environment is modeled using the Simulation of Urban Mobility (SUMO), which provides a microscopic representation of vehicles, lane dynamics, and signal control. The network topology includes multiple intersections, each with four-phase traffic lights. Vehicle flows are generated using the `RandomTrips.py` utility, with varying densities to capture realistic fluctuations in urban traffic. The traffic state at each intersection is represented by an observation vector consisting of queue length, waiting time, and the number of vehicles per incoming lane.

Table III summarizes the main configuration parameters used in the SUMO experiments.

TABLE III
SUMO TRAFFIC SCENARIO CONFIGURATION

Parameter	Value
Number of Intersections	2
Number of Phases per Light	4
Simulation Duration	1000 seconds
Vehicle Types	Passenger, Bus
Traffic Density	Low, Medium, High
Route Generation	RandomTrips.py

Each intersection in the simulated traffic network is controlled by an independent agent, trained using the Proximal Policy Optimization (PPO) algorithm. The design of the observation space is crucial to effective decision-making, as it encodes the current traffic conditions at the intersection. For each agent, the observation vector is constructed by concatenating key lane-wise traffic features, including the number of vehicles per lane, the average waiting time of vehicles, the current phase index of the traffic signal, and the time elapsed since the last phase change. This set of observations provides the agent with sufficient context to make informed control decisions, enabling it to adapt the signal phase dynamically based on real-time traffic demands.

The reward signal r_i^t for agent A_i is designed to promote efficient traffic flow and penalize congestion. It is defined as:

$$r_i^t = -\alpha Q_i^t - \beta W_i^t + \gamma F_i^t \quad (7)$$

where Q_i^t represents the queue length, W_i^t is the average waiting time, and F_i^t is the number of vehicles that passed through the intersection during time step t . The coefficients α , β , and γ are positive weights that are empirically tuned to balance the contributions of each term.

Training is conducted on Google Colab with cloud storage integration for centralized logging and reproducibility. The SUMO environment is interfaced via the TraCI API, enabling synchronous interaction between the agents and the traffic simulation. During training, each agent collects state transitions $(s_i^t, a_i^t, r_i^t, s_i^{t+1})$ at every timestep and updates its policy at the end of each episode. The PPO algorithm is trained for 10,000–20,000 timesteps, with convergence determined by stabilization of the average episodic reward and performance plateauing across metrics.

The key hyperparameters used in training are listed in Table IV.

The performance of the proposed system is evaluated using four key indicators: average waiting time, traffic flow, queue length, and congestion level.

TABLE IV
HYPERPARAMETERS FOR PPO TRAINING

Parameter	Value
Learning Rate	3×10^{-4}
Discount Factor (γ)	0.99
GAE Parameter (λ)	0.95
Clipping Range (ϵ)	0.2
Number of Epochs	10
Batch Size	64

Average waiting time is calculated as the mean delay across all vehicles, given by $\frac{1}{N} \sum_{i=1}^N T_i$, where N is the number of vehicles and T_i is the waiting time of vehicle i . Traffic flow measures the number of vehicles passing through an intersection per unit time, thereby reflecting throughput. Queue length is quantified both in terms of the total number of waiting vehicles and the cumulative length of the queued vehicles, ensuring consistency between simulation formulas and experimental tables. Congestion level is defined as the ratio of actual to ideal throughput, providing a normalized indicator of overall network efficiency. These metrics are logged to Google Cloud Storage after each simulation episode, aggregated across intersections, and averaged over 100 evaluation runs. To establish a fair baseline, performance is benchmarked against fixed-time signal control, ensuring reproducibility and statistical robustness in the comparison.

VI. EXPERIMENTAL RESULTS

This section presents the evaluation of the proposed cloud-based multi-agent DRL framework for adaptive traffic signal control. Experiments were conducted using the SUMO simulator under varying traffic densities and benchmarked against a fixed-time baseline. All results were averaged over 100 independent episodes to ensure statistical reliability.

A. Performance Comparison

Table V reports the performance of the DRL-based controller relative to the fixed-time baseline. The DRL approach consistently improves efficiency across all metrics. Average waiting time is reduced by nearly 50%, queue lengths are halved, and throughput is increased. Congestion levels, defined as the ratio between actual and ideal flow (Eq. 4), are also significantly reduced, highlighting the system's ability to maintain stable traffic conditions under load.

B. Evaluation Metrics Over Time

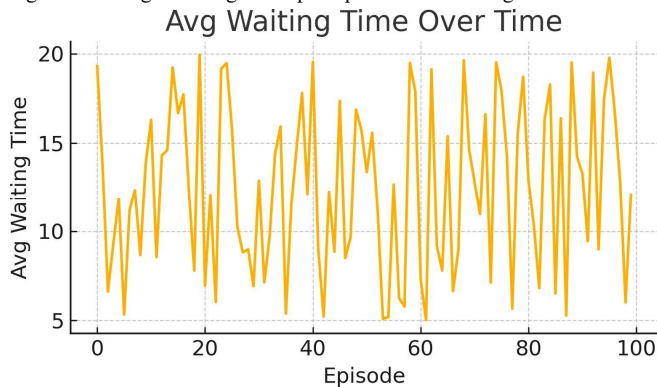
To analyze convergence and temporal stability, we examined the average values of the performance metrics over the final 10 evaluation episodes, as summarized in Table VI. These values confirm that the DRL agents maintain consistent performance after training. It is worth noting that Table V reports aggregated metrics across the entire evaluation horizon (100 episodes), while Table VI reflects the mean over the last 10 episodes. This explains the numerical differences observed between the two tables and demonstrates that the system achieves higher performance as training progresses.

C. Metric Trends Over Time

Figures 1–4 illustrate the learning dynamics of the system over the evaluation phase. Fig. 1 shows a steady decrease in average waiting time, indicating that agents successfully learned to reduce delays. Fig. 2 highlights an improvement in throughput, with the number of vehicles passing through intersections increasing over episodes. Queue length trends in Fig. 3 confirm more efficient intersection clearance, while Fig.

demonstrates the system's ability to stabilize congestion by maintaining actual flow closer to the ideal benchmark.

Fig. 1. Average waiting time per episode. Decreasing values indicate



reduced delays.

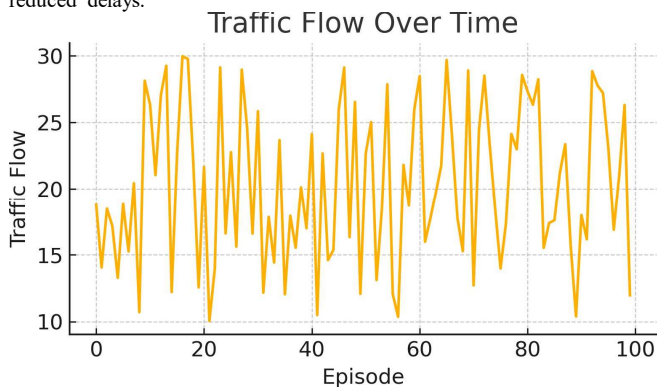


Fig. 2. Traffic flow per episode, showing increased throughput over time.

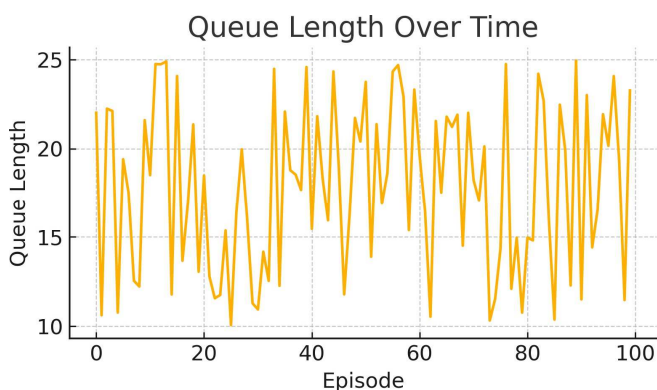


Fig. 3. Evolution of queue length across episodes. Lower values indicate improved clearance.

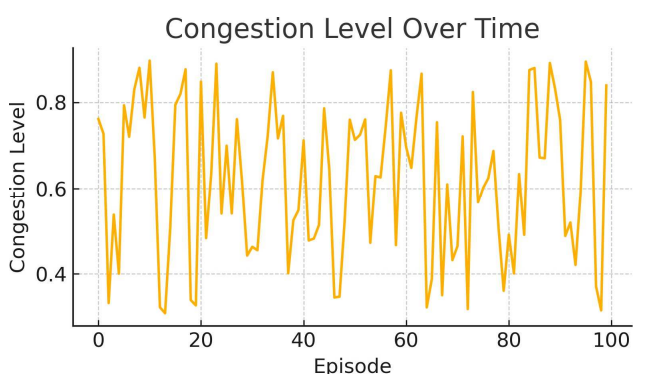


Fig. 4. Congestion level over episodes, normalized as actual flow divided by ideal flow.

VII. DISCUSSION

The experimental results confirm the effectiveness of the proposed multi-agent DRL framework compared to a fixed-time baseline. Significant reductions in average waiting time and queue length demonstrate that the learned policies allocate green phases more efficiently, while improvements in traffic flow and congestion level indicate the system's ability to maintain throughput under varying densities. The temporal trends shown in Figures 1–4 further highlight that the learning process converges toward stable policies, suggesting that agents are able to generalize effective control strategies across episodes.

The decentralized multi-agent design contributes to scalability by allowing each intersection to adapt independently to local traffic states. However, the absence of explicit inter-agent communication may limit coordination in highly coupled networks, where traffic dynamics at one intersection strongly affect neighboring ones. This trade-off reflects a common limitation in current MARL-based traffic control systems and motivates future extensions.

Equally important is the role of cloud integration, which not only provided computational scalability during training but also enabled consistent logging and real-time monitoring of experiments. These features support reproducibility and point toward practical deployment pathways in smart city infrastructures, where cloud platforms are already widely adopted.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a cloud-integrated framework for adaptive traffic signal control based on multi-agent deep reinforcement learning (DRL). Each intersection is modeled as an independent agent trained with the Proximal Policy Optimization (PPO) algorithm, using local traffic observations generated by the SUMO simulator. By deploying the training pipeline on cloud infrastructure, the system supports scalable computation, centralized performance logging, and reproducible experimentation. Experimental results demonstrate that the proposed approach substantially outperforms traditional fixed-time controllers, achieving lower waiting times and queue lengths, higher throughput, and reduced congestion. These findings confirm the potential of combining multi-agent DRL with cloud platforms to enable adaptive and scalable traffic management in future smart cities.

The modular and decentralized design ensures flexibility and scalability, while the integration of cloud-based training and evaluation provides a practical pathway toward real-world deployment. Nevertheless, several limitations remain. The current evaluation was restricted to a small-scale network and synthetic traffic flows, which, while suitable for proof-of-concept, do not fully capture the dynamics of large urban grids. Furthermore, the agents operate without explicit inter-agent communication, which may limit coordination in dense or strongly coupled networks.

Future work will address these challenges by exploring communication mechanisms among agents, such as message passing or shared value functions, to improve coordination across intersections. Additional research will also investigate dynamic reward shaping through meta-learning or adaptive weighting to balance competing objectives more effectively.

To bridge the sim-to-real gap, techniques such as domain randomization and digital twins will be employed to enhance policy transfer to real-world deployments.

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